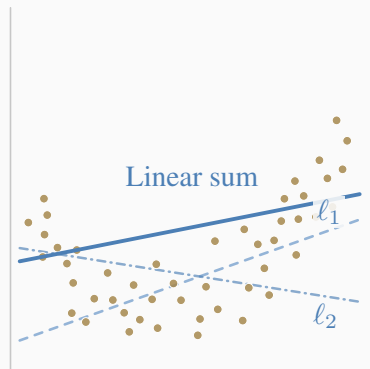


(a) Response spaces under a finite fast-state budget

More Linear State

State: $(r + 1)dd_v$



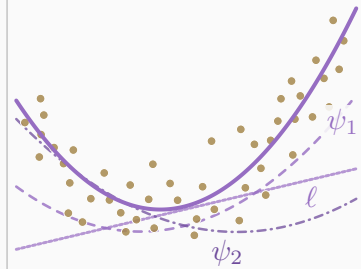
$$f = \ell_1 + \ell_2$$

Same linear response family
 $(r + 1)$ memory matrices

FVLA

State: $(r + 1)dd_v$

Linear backbone +
learned quadratic residual



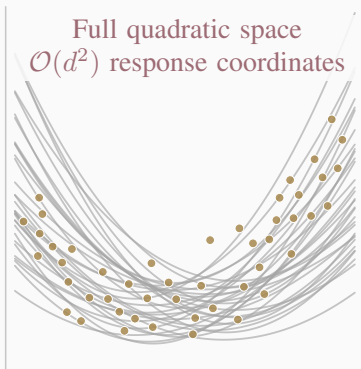
$$f = \ell + \psi_1 + \psi_2$$

Learned response subspace
Shared P_h + context-specific $R_t^{(h)}$

Full Quadratic

State: $\mathcal{O}(d^2 d_v)$

Full quadratic space
 $\mathcal{O}(d^2)$ response coordinates



All pairwise interactions
 $\mathcal{O}(d^2)$ feature coordinates

• Sample points

(b) One slot, two queries

History: *Alice gave Bob the book*

Query:

giver?

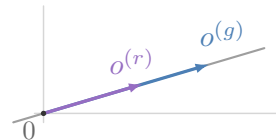
receiver?



latent slot i

Fixed-Value

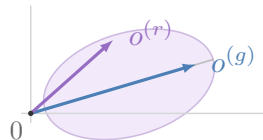
Through-origin line



Query-scaled
fixed value vector

Function-Valued

query-dependent response



Query-dependent
readout direction